Practice workbook · Explain heart and breathing changes in exercise and how carbon dioxide controls breathing

Raised heart and breathing rate, blood redirection, adrenaline, nerve control, recovery and oxygen debt, and chemoreceptors and the medulla responding to carbon dioxide.

11 question parts: 1 worked-example retry and 10 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-EXERCISE-AND-BREATHING-CONTROL

1. Question 5 · part b.i · 4 marks

June 2023 Paper 2

Worked-example retry · Attempt part b.i only.

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5 (a) A red blood cell travelling from the liver to the kidney passes through six different arteries and veins.

Complete the flow chart by naming these blood vessels in order. The first one has been done for you.

(5)

LIVER

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hepatic vein

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KIDNEY

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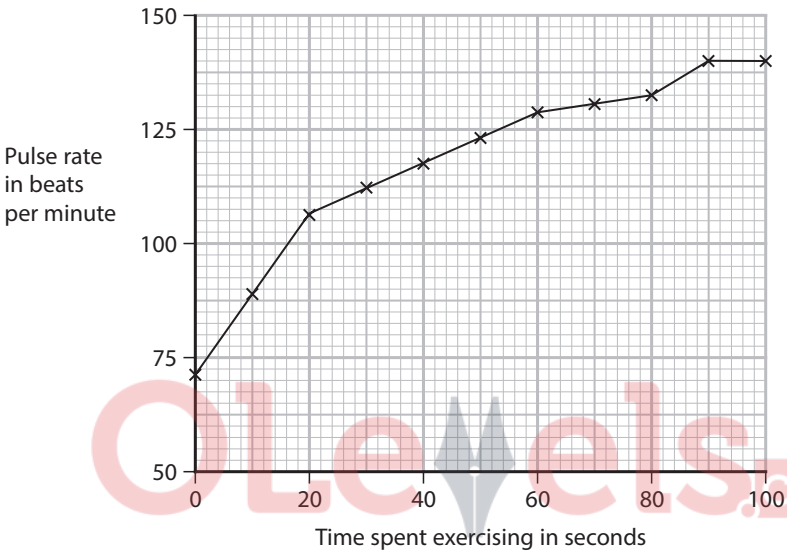
1. Question 5 · part b.i · 4 marks

June 2023 Paper 2

Worked-example retry · Attempt part b.i only.

(b) A student investigates the effect of time spent exercising on pulse rate.

The graph shows the student's results.



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1. Question 5 · part b.i · 4 marks

June 2023 Paper 2

Worked-example retry · Attempt part b.i only.

(i) Explain the change in pulse rate recorded by the student.

(4)

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1. Question 5 · part b.i · 4 marks

June 2023 Paper 2

Worked-example retry · Attempt part b.i only.

(ii) Describe how the student could carry out the investigation.

(6)

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(Total for Question 5 = 15 marks)

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1. Question 5 · part b.i · 4 marks · Working and answer

Working and answer

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2. Question 8 · part c · 4 marks

June 2019 Paper 1R

Further practice · Attempt part c only.

8 The table shows the rate of blood flow through different organs when a person is at rest. The table also shows the masses of these organs.

Organ	Mass in kg	Rate of blood flow in cm^3 per minute	
		through the whole organ	per 100g of the organ
kidneys	0.3	1200	
skeletal muscles	25.0		3
liver	1.5	1500	100

(a) Calculate the missing values and put them in the table.

(4)

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(b) Explain why the liver has a high rate of blood flow.

(4)

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2. Question 8 · part c · 4 marks

June 2019 Paper 1R

Further practice · Attempt part c only.

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(c) Explain any changes that would occur in the blood flow through these organs during vigorous exercise.

(4)

(Total for Question 8 = 12 marks)

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Working and answer

3. Question 9 · part a.ii · 4 marks

January 2020 Paper 1

Further practice · Attempt part a.ii only.

9 In sports such as archery and shooting, beta-blockers are used illegally by some competitors to help reduce the effects of stress and anxiety, which can affect performance.

(a) (i) Suggest why the use of beta-blockers in competitive sport is illegal.

(1)

(ii) Stress and anxiety can cause an increase in the release of adrenaline.

Explain the effects of adrenaline on the body.

(4)

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3. Question 9 · part a.ii · 4 marks

January 2020 Paper 1

Further practice · Attempt part a.ii only.

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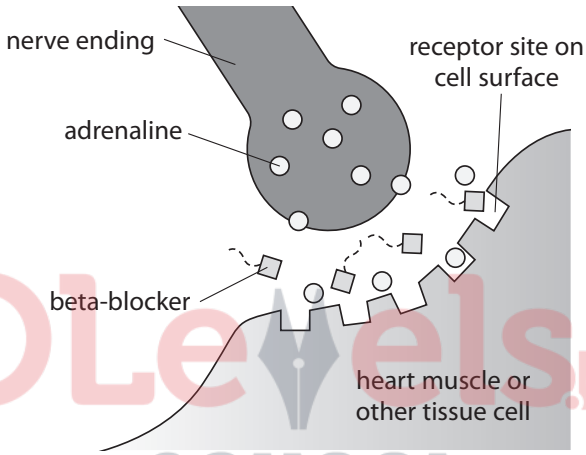
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(b) Adrenaline acts as a hormone and as a neurotransmitter.

Beta-blockers act at the junction between nerve endings and body tissues where adrenaline is found.

The diagram shows the action of beta-blockers at a synapse.



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(i) Explain how beta-blockers reduce symptoms of stress and anxiety.

(3)

(ii) Suggest how the action of beta-blockers would be a disadvantage in active sports such as athletics.

(2)

(Total for Question 9 = 10 marks)

3. Question 9 · part a.ii · 4 marks · Working and answer

Working and answer

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4. Question 2 · part d · 1 mark

January 2020 Paper 2

Further practice · Attempt part d only.

2 A student records her pulse rate in beats per minute (bpm) after different activities.

These are the student's results.

- at rest 78 bpm
- walking 95 bpm
- jogging 138 bpm
- running 146 bpm

(a) Draw a table to show the student's results.

(3)

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4. Question 2 · part d · 1 mark

January 2020 Paper 2

Further practice · Attempt part d only.

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(b) Describe a method that the student could use to obtain these results.

(4)

(c) State one factor that the student should control during the investigation.

(1)

(d) Name a body process, other than pulse rate, that increases during exercise.

(1)

(Total for Question 2 = 9 marks)

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4. Question 2 · part d · 1 mark · Working and answer

Working and answer

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5. Question 5 · part c · 2 marks

November 2020 Paper 2R

Further practice · Attempt part c only.

5 A scientist investigates how the drug, atropine, affects the heart rate in humans.

Atropine paralyses the vagus nerve which carries impulses from the brain to the heart.

The heart rates of ten people are measured at intervals of two minutes for eight minutes, after which, atropine is injected.

The scientist continues to measure the heart rates.

The scientist's results are shown in the table.

Time in minutes	Heart rate in beats per minute
0	90
2	81
4	78
6	80
8	82
atropine injected → 9	100
10	120
11	148
12	160
13	138
15	128
17	120
19	116
21	109
23	100

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5. Question 5 · part c · 2 marks

November 2020 Paper 2R

Further practice · Attempt part c only.

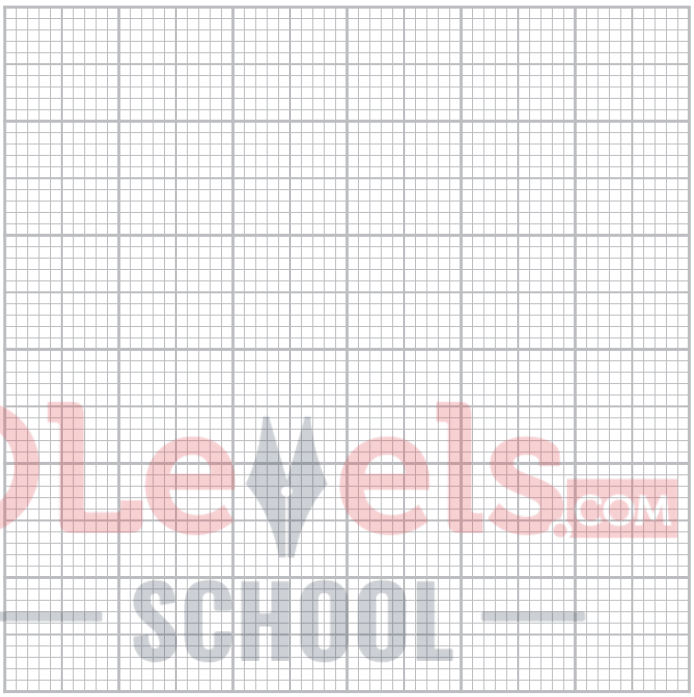
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(a) (i) Plot the scientist's results on the grid. Draw a curve of best fit.

(6)



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(ii) Explain why ten people are injected with atropine rather than just one person.

(2)

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.....

(b) (i) Calculate the percentage change in the heart rate at 12 minutes compared to the heart rate at the start of the investigation.

(2)

percentage change = ...%

5. Question 5 · part c · 2 marks

November 2020 Paper 2R

Further practice · Attempt part c only.

(ii) Explain why this percentage change is not achieved as soon as atropine is injected.

(2)

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(c) Suggest the function of the vagus nerve.

(2)

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(Total for Question 5 = 14 marks)

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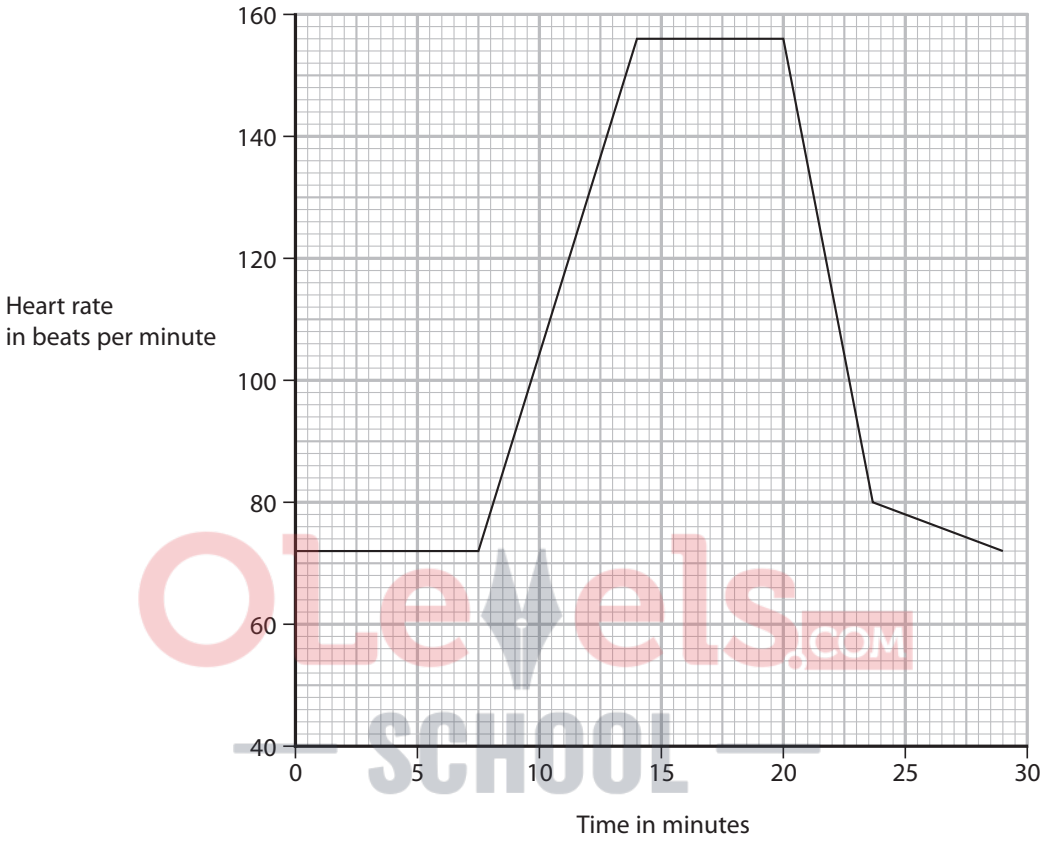
Working and answer

6. Question 2 · part a.ii · 1 mark

June 2022 Paper 1

Further practice · Attempt part a.ii only.

2 (a) The graph shows how a person's heart rate changes with exercise.



(i) Determine the difference between the maximum heart rate and the resting heart rate.

(2)

difference = ... beats per minute

(ii) Which of these is the time taken for this person to recover from the exercise?

(1)

- ☒ **A** 4 minutes
- ☒ **B** 6 minutes
- ☒ **C** 9 minutes
- ☒ **D** 15 minutes

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6. Question 2 · part a.ii · 1 mark

June 2022 Paper 1

Further practice · Attempt part a.ii only.

(iii) Complete the passage about what happens in the body during exercise using the correct words.

(4)

During exercise ..... gas moves from the lungs into the blood by a process called ..... . This gas is needed by cells for ..... respiration which releases energy. Also during exercise, the amount of urine produced ..... as water is lost in sweat.

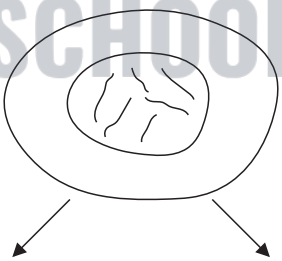
(b) (i) During exercise, body cells get damaged.

Body cells are replaced by a process called mitosis.

The diagram represents one body cell but shows only 6 of the 46 chromosomes.

Complete the diagram to show the new body cells produced from this cell by mitosis.

(2)



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6. Question 2 · part a.ii · 1 mark · Working and answer

Working and answer

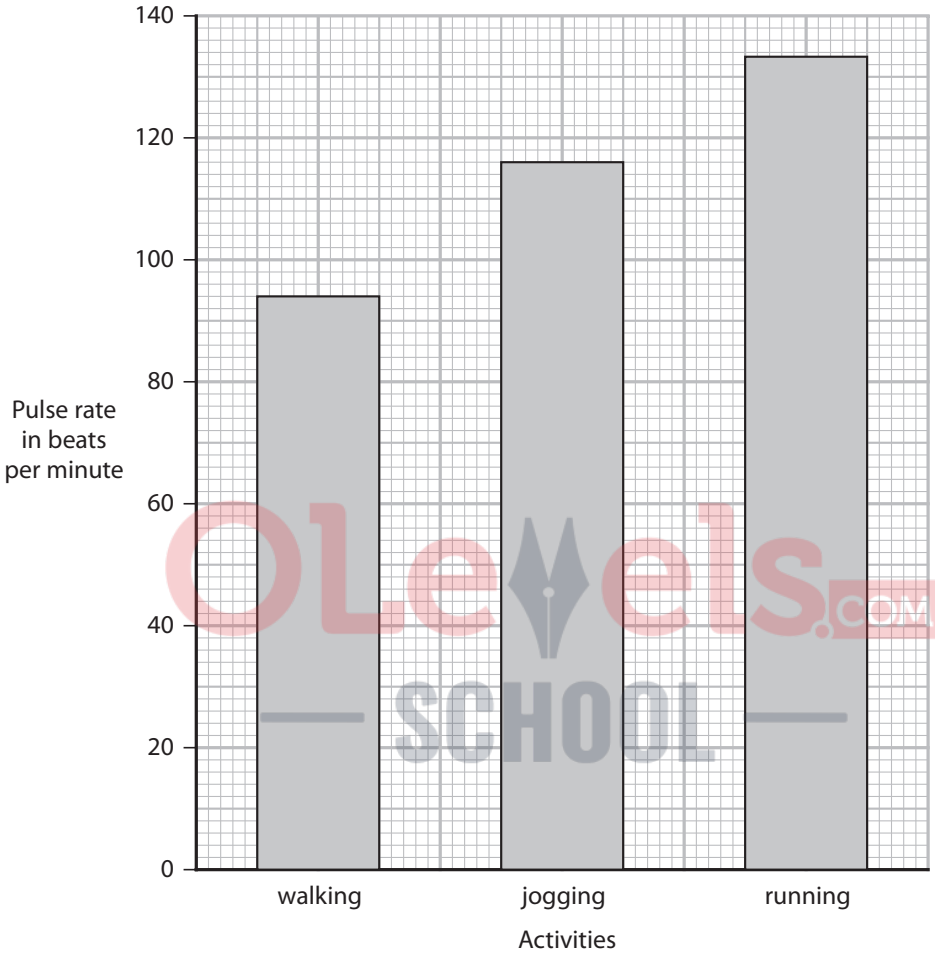
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7. Question 4 · part b · 4 marks

June 2023 Paper 2R

Further practice · Attempt part b only.

4 (a) The bar chart shows the pulse rate of a student at the end of three different activities.



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7. Question 4 · part b · 4 marks

June 2023 Paper 2R

Further practice · Attempt part b only.

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(i) Draw a table that the student could use to record these results.

(3)

(ii) Complete the table by adding the student's results.

(2)

(iii) Describe a method that the student could use to obtain the results shown in the bar chart.

(4)

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7. Question 4 · part b · 4 marks

June 2023 Paper 2R

Further practice · Attempt part b only.

(b) The pulse rates at the end of walking and running are different.

Explain this difference.

(4)

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(Total for Question 4 = 13 marks)

7. Question 4 · part b · 4 marks · Working and answer

Working and answer

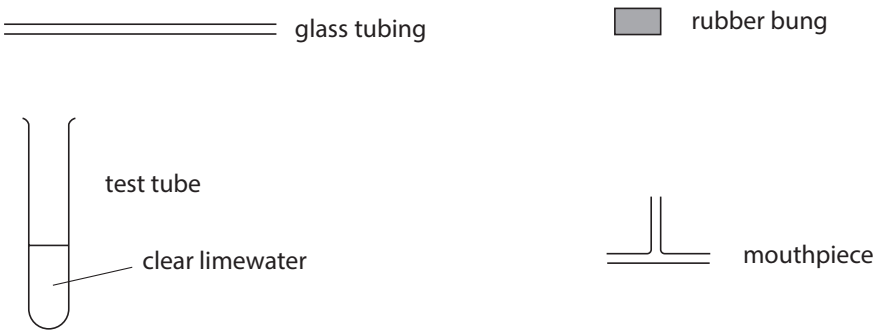
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8. Question 4 · part b.ii · 4 marks

June 2024 Paper 1

Further practice · Attempt part b.ii only.

4 (a) The diagram shows some scientific equipment.



(i) Draw a diagram to show how this equipment is set up to investigate the difference between the carbon dioxide concentration of inspired air and of expired air.

(3)

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8. Question 4 · part b.ii · 4 marks

June 2024 Paper 1

Further practice · Attempt part b.ii only.

(ii) State one safety precaution in this investigation.

(1)

(b) A student investigates their breathing rate at rest, after gentle exercise and again after vigorous exercise.

The student repeats each exercise three times.

The student's mean results are shown in the table.

Mean breathing rate in breaths per minute		
at rest	**after gentle exercise**	**after vigorous exercise**
14	19	50

(i) Give one control variable in this investigation.

(1)

(ii) Explain how the breathing rate after vigorous exercise benefits working muscles.

(4)

(Total for Question 4 = 9 marks)

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8. Question 4 · part b.ii · 4 marks · Working and answer

Working and answer

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9. Question 2 · part b · 1 mark

January 2022 Paper 2

Further practice · Attempt part b only.

2 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

The breathing system consists of airways that allow the movement of air into the lungs. Air can be breathed in through the nose or through the mouth. The lining of the nose has a good supply of blood capillaries and a layer of sticky mucus.

The whole of the trachea and the larger air passages have C-shaped rings of cartilage within their walls. These rings help the air to flow freely into the lungs. The trachea divides into two tubes which then divide further into finer branches. These finer branches eventually end in air sacs. It is in the air sacs that gaseous exchange occurs.

The lungs are located in the thorax which is surrounded by the ribs. The thorax is separated from the abdomen by the diaphragm. Together, the ribs and the diaphragm can change the volume of the thorax and therefore the pressure in the thorax.

During breathing, at rest, an average of 0.50 dm^3 of air is breathed in and out. The air breathed in contains 20% oxygen and the air breathed out contains 16% oxygen. At rest, an adult will breathe 17 times a minute on average. The breathing rate is controlled by a part of the brain which receives information about carbon dioxide levels in the blood.

Carbon monoxide may be present in the air that is breathed in. The carbon monoxide can enter the blood and combine with haemoglobin in the red blood cells. Carbon monoxide is found in cigarette smoke and car exhaust fumes.

(a) (i) Discuss whether it is better to breathe in through the nose or the mouth. (lines 2 to 3)

(3)

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9. Question 2 · part b · 1 mark

January 2022 Paper 2

Further practice · Attempt part b only.

(ii) Explain why the larger air passages have C-shaped rings of cartilage within their walls. (lines 4 to 5)

(2)

(iii) The trachea divides into two tubes.

What is the name of these tubes? (line 6)

(1)

- ☒ **A** alveoli
- ☒ **B** bronchi
- ☒ **C** bronchioles
- ☒ **D** oesophagus

(b) Which of these is the name of the part of the brain that controls breathing rate? (line 16)

(1)

- ☒ **A** cerebrum
- ☒ **B** cerebellum
- ☒ **C** hypothalamus
- ☒ **D** medulla oblongata

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9. Question 2 · part b · 1 mark

January 2022 Paper 2

Further practice · Attempt part b only.

(c) Calculate the average volume of oxygen absorbed by the lungs during one minute of breathing, at rest. (lines 13 to 15)

(3)

volume of oxygen absorbed = ... dm^3

(d) Explain why pregnant women are advised not to smoke cigarettes.

(4)

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(Total for Question 2 = 14 marks)

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9. Question 2 · part b · 1 mark · Working and answer

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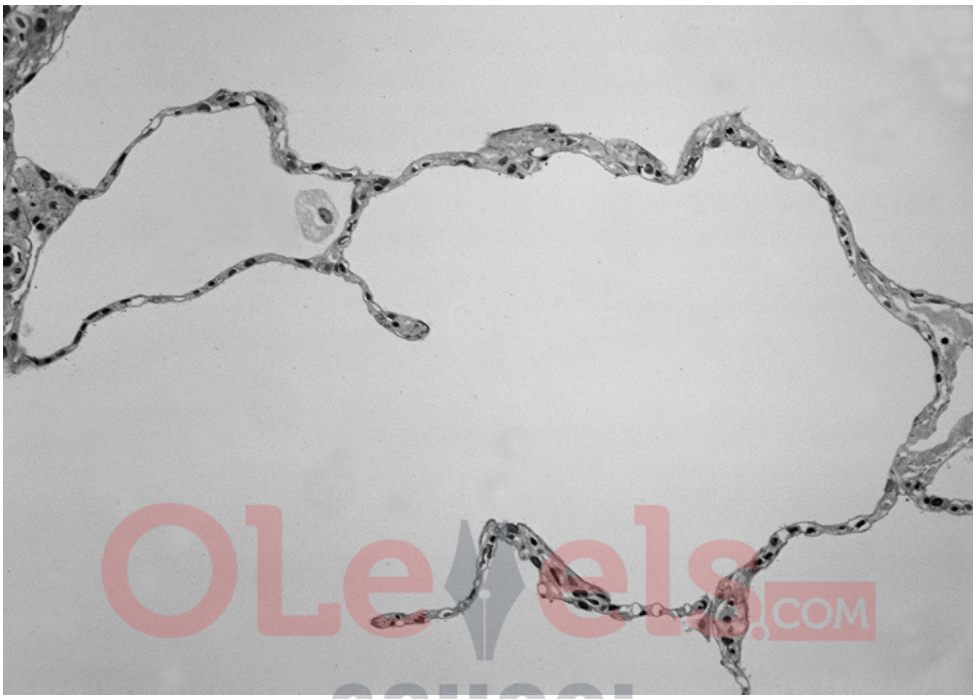
10. Question 2 · part b.iv · 4 marks

June 2023 Paper 1

Further practice · Attempt part b.iv only.

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2 The photograph shows alveoli in the lungs. Alveoli are the site of exchange of gasses between the air and the blood.



(Source: © Ed Reschke/Getty Images)

(a) (i) State, in the correct order, the structures that air must pass through from the atmosphere to the alveoli.

(2)

.....

.....

(ii) State two features, shown in the diagram, that allow the efficient exchange of gases.

(2)

1 .....

.....

2 .....

.....

10. Question 2 · part b.iv · 4 marks

June 2023 Paper 1

Further practice · Attempt part b.iv only.

(b) (i) Which is the percentage of carbon dioxide in the atmosphere?

(1)

- ☒ **A** 0.01
- ☒ **B** 0.04
- ☒ **C** 0.10
- ☒ **D** 0.50

(ii) The table shows the results obtained when a person at rest breathes in air that contains different percentages of carbon dioxide.

Percentage (%) carbon dioxide in air	0.02	1.60	3.00	6.00
Tidal volume in cm^3	500	700	1200	2100
Rate of breathing in breaths per minute	14	15	16	28

Calculate the volume of air in dm^3 breathed in per minute when the person breathes air containing 3.00% carbon dioxide.

[1 dm^3 = 1000 cm^3]

(2)

volume of air = ... dm^3

(iii) The air a person breathes in contains 20% oxygen. The body uses 20% of this oxygen.

Calculate the volume of oxygen used per minute when the person breathes air containing 1.60% carbon dioxide.

(3)

volume of oxygen = ... cm^3

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10. Question 2 · part b.iv · 4 marks

June 2023 Paper 1

Further practice · Attempt part b.iv only.

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(iv) Explain the effect of increasing the percentage of carbon dioxide in the air a person breathes in on the tidal volume and rate of breathing.

(4)

(Total for Question 2 = 14 marks)

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Working and answer

11. Question 7 · part b · 2 marks

June 2024 Paper 1

Further practice · Attempt part b only.

7 (a) The list gives some structures found in the thorax.

- intercostal muscles
- ribs
- diaphragm

Describe how these structures work together during inhalation.

(4)

(b) Describe the role of chemoreceptors in the carotid arteries.

(2)

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11. Question 7 · part b · 2 marks

June 2024 Paper 1

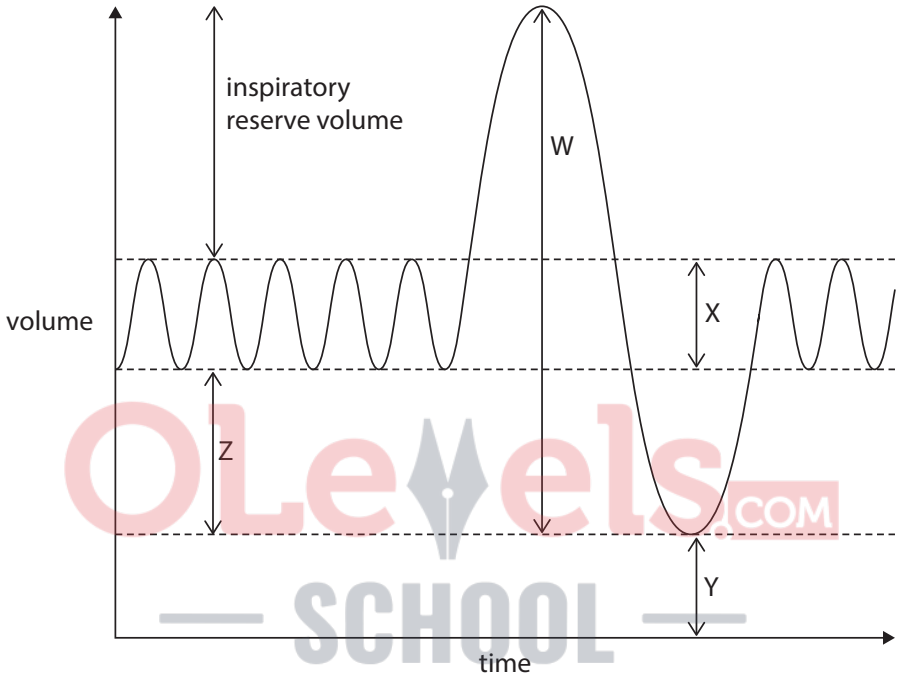
Further practice · Attempt part b only.

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(c) The graph shows different lung volumes.



(Source: https://www.inds.co.uk/education/how2/measure-lung-volume/)

(i) Give the name of the lung volume represented by letter X.

(1)

.....

(ii) Give the name of the lung volume represented by letter Y.

(1)

.....

(iii) Give the name of the lung volume represented by letter W.

(1)

.....

(Total for Question 7 = 9 marks)

11. Question 7 · part b · 2 marks · Working and answer

Working and answer

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